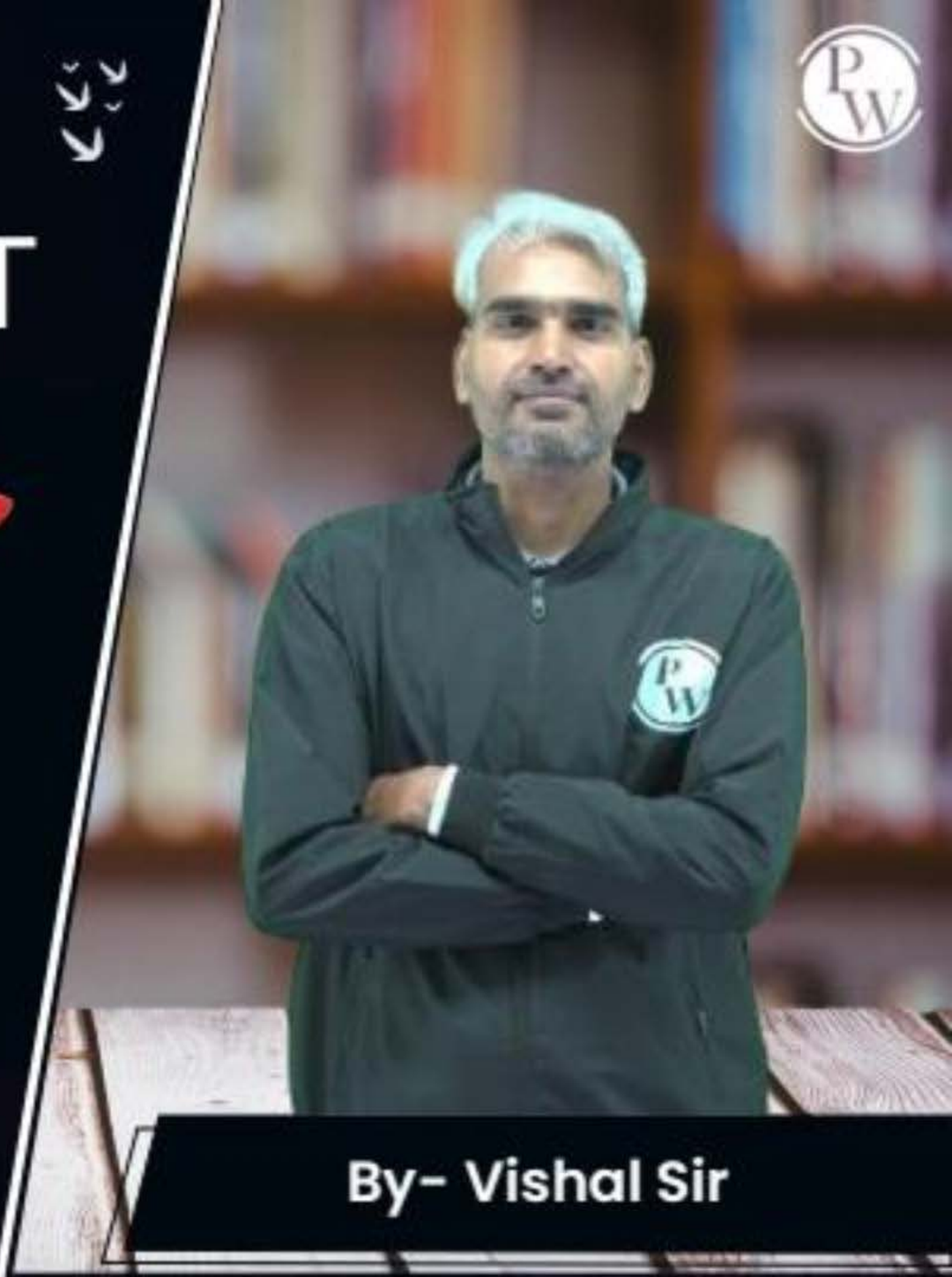
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 13



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Recap of Previous Lecture



- **Topic** Lossless join decomposition
- **Topic** Projection, Selection, and Cross product
- **Topic** Natural join

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Topics to be Covered



- Topic: Lossless Natural Join
- Topic: Normal forms
- Topic: First normal form (1NF)

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Lossless Natural Join.

* Consider the following relational table.

Case (1) If relation R is decomposed into two subrelations $R_1(AB)$ & $R_2(BC)$, then check whether the decomposition is lossless Join decomposition or not?

R

A	B	C
1	1	1
2	1	2
3	2	1

R_1

A	B
1	1
2	1
3	2

R_2

B	C
1	1
1	2
2	1

$R_1 \bowtie R_2$

A	B	C
1	1	1
1	1	2
2	1	1
2	1	2
3	2	1

tuple from R

Extra tuples { Known as Spurious tuples }

* Common attribute between R_1 & R_2 is B
Values of B are neither unique in R_1 nor unique in R_2
i.e. B can not be the super key of any of the two relations
And Natural join of R_1 & R_2 is lossy

$\therefore R_1 \bowtie R_2 \supset R$

Hence, Lossy decomposition

H.W.

Consider the following relational table.

R

A	B	C
1	1	1
2	1	2
3	2	1

② If relation R is decomposed into two subrelations $R_1(AB)$ & $R_2(AC)$, then check whether the decomposition is lossless Join decomposition or not?

R_1=

A	B
1	1
2	1
3	2

R_2=

A	C
1	1
2	2
3	1

$R_1 \bowtie R_2$=

A	B	C
1	1	1
2	1	2
3	2	1

$R_1 \bowtie R_2 = R$

∴ Lossless Join decomposition

* Common attribute b/w R_1 & R_2 is A.
* Values of 'A' are unique in R_1 as well as in R_2

and Natural join of R_1 & R_2 is lossless

H.W.

Consider the following relational table.

R

A	B	C
1	1	2
2	1	2
3	2	1

③ If relation R is decomposed into two subrelations $R_1(AB)$ & $R_2(BC)$, then check whether the decomposition is lossless Join decomposition or not?

R_1=

A	B
1	1
2	1
3	2

R_2=

B	C
1	2
1	1
2	1

$R_1 \bowtie R_2$=

A	B	C
1	1	2
2	1	2
3	2	1

$R_1 \bowtie R_2 = R$

∴ Losless Join decomposition

* Common attributes between R_1 & R_2 is B.
* Values of B are not unique in R_1 but values of B are unique in R_2 and, Natural join of R_1 & R_2 is Lossless.



Topic : NOTE

* If Relation R is decomposed into two subrelations R_1 & R_2, then this decomposition is lossless join decomposition if and only following three conditions holds true.

① Attributes of R_1 $\cup$ Attributes of R_2 = Attributes of R

② Attributes of R_1 $\cap$ Attributes of R_2 $\neq \emptyset$

③ Common attributes must be present and Common attributes must be a super key of at least one of the two relations.

{ Because if there are no common attributes, then Natural join will degenerate into Cross Product {i.e; $R_1 \bowtie R_2 = R_1 \times R_2$} and it will be lossy.

Q:-

#Q. Let R (A, B, C, D, E) be the relational schema with following FD set

F = { AB→C, C→D, B→E)

Which of the following decomposition is/are lossless join decomposition?

(i){R1(ABC), R2(CD)} → Attributes of R_1 ∪ Attributes of R_2 ≠ Attributes of R
∴ lossy

(ii){R1(ABC), R2(DE)} → Attributes of R_1 ∪ Attributes of R_2 = Attributes of R ✓
→ Attributes of R_1 ∩ Attributes of R_2 = {A,B,C} ∩ {D,E} = ∅
∴ lossy

(iii){R1(ABC), R2(CDE)} → Attributes of R_1 ∪ Attributes of R_2 = Attributes of R ✓
→ Attributes of R_1 ∩ Attributes of R_2 = {C} ≠ ∅ ✓
→ Common attributes = {C} ⇒ $\{C\}^+$ w.r.t. = {C,D}
{3rd Condn is dis-satisfied}
Not all attributes of R_1 ∴ Not a S.K. of R_1
Not all attributes of R_2 ∴ Not a S.K. of R_2

(iv){R1(ABCD), R2(BE)} ✓
Attribute-wise $R_1 \cup R_2 = R$ ✓
Attribute-wise $R_1 \cap R = \{B\} \neq \emptyset$ ✓
$(B)^+$ w.r.t. F = {B,E} → All attributes of R_2 ∴ S.K. of R_2 ✓

Note :- ① $R_1 \bowtie R_2 = R_2 \bowtie R_1$ { datawise }

{ i.e. Natural join is Commutative }

② $(R_1 \bowtie R_2) \bowtie R_3 = R_1 \bowtie (R_2 \bowtie R_3)$ { datawise }

{ i.e. Natural join is associative }

#Q. Let R (A, B, C, D, E, F) be the relational schema with following FD set

F = { AB→C, BC→A, AC→B, AD→E, B→D, E→F)

Which of the following decomposition is lossless join decomposition.

(i) {R1(ABC), R2(ADF), R3(ACDE)}

(ii) {R1(ABC), R2(ABDE), R3(EF)}

(iii){R1(AB), R2(BC), R3(ABDE), R4(EF)}

$(R_2 \bowtie R_2)(BCD)$

$R_4(BD)$

$(B)^+ = \{B, D\}$

$\Rightarrow R_2(BC)$ can not be joined with other relations or join of other relations in any order s.t join is lossless

∴ It is lossy join decomposition

(i) $R_1(ABC)$ $R_2(ADF)$ $R_3(ACDE)$

$\cap = (A)^+ = \{A\}$

Neither a S.k of R_1
Nor a S.k. of R_2
∴ Don't join in this order

$\cap = (AC)^+ = \{A, C, B, D, E, F\}$

'AC' is S.k of both R_1 as well as R_3
∴ We can join R_1 & R_3

After natural join of R_1 & R_3, Attributes of R_1 & R_3 Will be combined, i.e; $(R_1 \bowtie R_3)(A, B, C, D, E)$

$\cap = (AD)^+ = \{A, D, E, F\}$

All attributes of R_2
∴ S.k. of R_2

Hence we can join $(R_1 \bowtie R_3)$ with R_2

$((R_1 \bowtie R_3) \bowtie R_2)(A, B, C, D, E, F)$

If there exists any order in which relations can be joined, s.t. join is lossless at every point of join, then overall decomposition is lossless join decomposition

(ii) $R_1(ABC)$ $R_2(ABDE)$ $R_3(EF)$

$\cap = (AB)^+ = \{A, B, C, D, E, F\}$

Sik. of both the relation

∴ We can join

$(R_1 \bowtie R_2)(ABCDE)$

$\cap = (E)^+ = \{E, F\}$

Sik. of R_3

∴ We can join

$((R_1 \bowtie R_2) \bowtie R_3)(A, B, C, D, E, F)$ ✓

Hence lossless join decomposition.



Topic : Normalization

- Normalization is the process of decomposing the relation into sub-relations, such that redundancy is reduced or eliminated.
- To specify the different types of redundancy we use the concept of "Normal Forms"



Topic : Normal forms

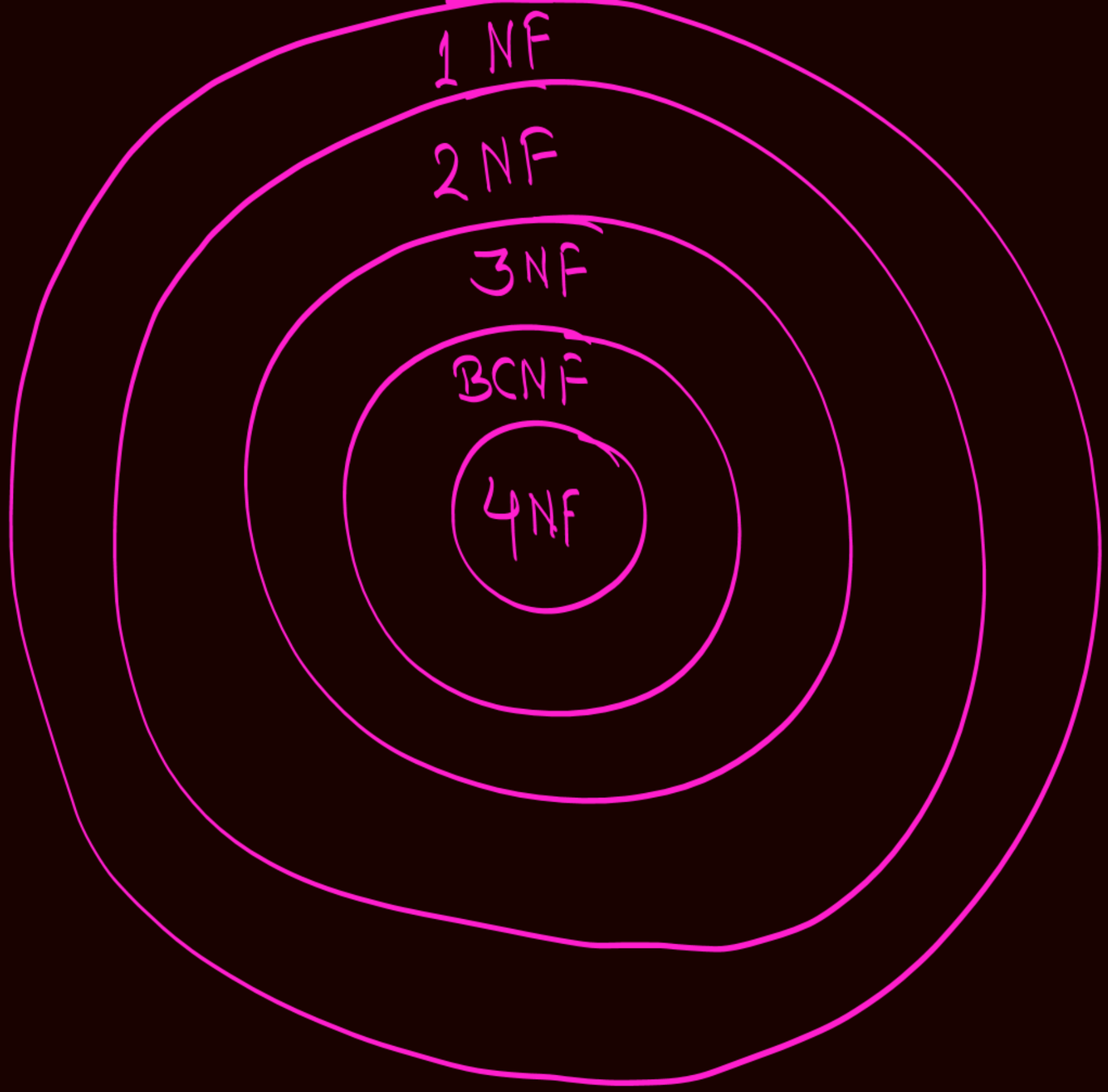
Boyce Codd Normal form → introduced the Concept of relational model.

There are various types of normal forms,

1 NF | 2 NF | 3NF | BCNF | 4NF

* Upto BCNF we will try to eliminate the redundancies present in the relation because of functional dependencies.
* If a relation is in BCNF, then there will be no redundancy in that relation because of functional dependencies, but a relation which is in BCNF may still suffer from redundancy present in it because of Multi-valued dependencies (MVD)

* 4NF is related to multi-valued dependencies
* In 4NF we will try to eliminate the redundancies present in the relation because of multi-valued dependency!


1 NF
2NF
3NF
BCNF
4NF



2 mins Summary

- Topic: Lossless Natural Join
- Topic: Normal forms
- Topic: First normal form (1NF)

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THANK - YOU

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